Question 103 | Question bank page 10 | Q.2

Create a Student Record Management PWA.

Project type: Progressive Web Application

Steps to perform

1. Create an ASP.NET Core MVC project or a suitable static web project in Visual Studio.

2. Add HTML or Razor pages, manifest.json, service-worker.js, and any referenced icons under wwwroot.

3. Register the service worker and link the manifest in the HTML layout.

4. Run the site on localhost or HTTPS and inspect it in a modern browser.

5. Use browser Developer Tools to check manifest, service worker, cache and offline operation. Push notifications require additional permission and subscription/server setup.

Closest supplied practical: </body> </html> Create a Student Record Management PWA.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create an MVC project named StudentRecordPWA. Add the student model and view, then
add the manifest and service worker for the Students page.
Filename: index.cshtml    Location: Views/Students/Index.cshtml
@model IEnumerable<StudentRecordPWA.Models.Students>
<h2>Student Records</h2>
<table border="1" cellpadding="10">
    <tr>
        <th>Id</th>
        <th>Name</th>
        <th>Course</th>
        <th>Age</th>
    </tr>
    @foreach (var item in Model)
    {
        <tr>
            <td>@item.Id</td>
            <td>@item.Name</td>
            <td>@item.Course</td>
            <td>@item.Age</td>
        </tr>
    }
</table>
Filename: manifest.json    Location: wwwroot/manifest.json
{
  "name": "Student Record PWA",
  "short_name": "StudentPWA",
  "start_url": "/Students",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
Filename: Service-Worker    Location: wwwroot/service-worker.js
const CACHE_NAME = "student-pwa-cache";
const urlsToCache = [
    "/Students"
];
self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
```

```
});
self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
Filename: Layout    Location: Project file location not specified in source
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - StudentRecordPWA</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/StudentRecordPWA.styles.css" asp-append-version="true" />
    <link rel="manifest" href="/manifest.json">
</head>

<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white bor
der-bottom box-shadow
mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">StudentRecordPWA</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse"
aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>
    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
    </div>
    <footer class="border-top footer text-muted">
        <div class="container">
            &copy; 2026 - StudentRecordPWA - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
        </div>
    </footer>
    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>
    @await RenderSectionAsync("Scripts", required: false)
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'));
        }
    </script>
</body>
</html>
Filename: StudentController.cs    Location: Project file location not specified in source
using Microsoft.AspNetCore.Mvc;
using StudentRecordPWA.Models;
using System.Collections.Generic;
namespace StudentRecordPWA.Controllers
{
    public class StudentsController : Controller
    {
```

```
        public IActionResult Index()
        {
            List<Students> students = new List<Students>()
            {
                new S
tudents { Id = 1, Name = "Rahul", Course = "BSc IT", Age = 20 },
                new Students { Id = 2, Name = "Amit", Course = "BCA", Age = 21 },
                new Students { Id = 3, Name = "Sneha", Course = "BSc CS", Age = 19 }
            };
            return View(students);
        }
    }
}
Filename: Students.cs    Location: Models/Students.cs
namespace StudentRecordPWA.Models
{
    public class Students
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Course { get; set; }
        public int Age { get; set; }
    }
}

Create an HTML page that would be the starting point of the application. This HTML
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.